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DATA SCIENCE

DAY 2

Building the Retrieval Pipeline

Using LlamaIndex, OpenAI Embeddings, and Pinecone to make enterprise HR knowledge searchable

Day 1
Clarity
Use case • RAG basics • Architecture

Day 2
Retrieval
LlamaIndex • Embeddings • Pinecone

Day 3
Application
LLM answers • Gradio • Demo

Mohammad Arshad • Decoding Data Science

From Day 1 to Day 2

From blueprint to working retrieval system

AI Accelerator Bootcamp

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Flow from Day 1 to Day 2

Keep the learning journey clean and sequential

Day 1 — What participants built mentally

- Defined the HR chatbot use case and target user
- Understood why enterprise AI needs RAG
- Mapped the end-to-end architecture
- Connected the CPE framework to the bootcamp build

Bridge

- Now the question becomes: how does the chatbot find the right company knowledge before it answers?
- Day 2 focuses on the retrieval engine — not just model prompting.

Day 2 — What gets built today

- Process HR documents with LlamaIndex
- Generate embeddings
- Store vectors in Pinecone
- Retrieve relevant context for user questions

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Day 2 learning outcomes

What participants should understand and build by the end of the session

Understand

- What LlamaIndex does in a RAG workflow
- How embeddings enable semantic search
- Why Pinecone is used as the vector database

Build

- A searchable HR knowledge base
- A working retrieval pipeline
- Sample query tests against indexed content

Prepare for Day 3

- Grounded context retrieval before answer generation
- A clean handoff to LLM response generation
- A stronger final chatbot demo in Gradio

Where Day 2 fits in the full system

Today sits in the middle of the enterprise AI stack

DAY 2 FOCUS

HR Documents → LlamaIndex Processing → Embeddings → Pinecone → Retrieval → LLM Response → Gradio UI

Raw enterprise knowledge becomes retrievable context here.

Simple mental model for beginners

Use this slide to reduce technical overload

LlamaIndex

- Organizes the knowledge workflow
- Loads documents and manages retrieval plumbing

Embeddings

- Turn text meaning into vectors
- Enable semantic similarity search

Pinecone

- Stores and searches vectors fast
- Returns relevant content for a query

Retrieval

- Finds the best context
- Feeds the LLM only what matters

Key message: we are not manually building every low-level step — LlamaIndex handles much of the document processing workflow so learners can focus on understanding the full system.

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Why retrieval matters in enterprise AI

The model sounds smart only when the knowledge layer is strong

Without retrieval

- Answers become generic
- Internal HR policy questions are unreliable
- Hallucinations increase
- User trust drops quickly

With retrieval

- Responses are grounded in enterprise documents
- The chatbot searches the right knowledge first
- Answers become more relevant and defensible
- The system starts to resemble a real enterprise assistant

Day 2 workflow blocks

A clean sequence for the live session

1	**1. Load documents**	Bring HR policy files into the workflow
2	**2. LlamaIndex processing**	Parse and organize content into retrievable units
3	**3. Generate embeddings**	Create semantic representations of the content
4	**4. Store in Pinecone**	Build the searchable vector index
5	**5. Retrieval test**	Run user questions and inspect returned context

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Live build plan

What participants will actually do during the session

Part A — Setup the knowledge base

- Inspect the HR documents
- Load them into LlamaIndex
- Understand what the framework is doing behind the scenes

Part B — Index the content

- Generate embeddings
- Connect to Pinecone
- Store the vectorized knowledge base

Part C — Test retrieval

- Ask realistic HR questions
- Inspect the retrieved context
- Check whether the system is finding the right information

How to judge retrieval quality

Day 2 is successful only if the right context comes back

Ask these questions

- Did the system retrieve the correct HR policy section?
- Did it miss an important section?
- Is there too much irrelevant noise?
- Would this context help the LLM answer correctly tomorrow?

Common mistakes

- Assuming the LLM will fix weak retrieval
- Testing only easy or obvious queries
- Ignoring the quality of source documents
- Treating RAG as upload-a-PDF-and-magic-happens

Expected outcome of Day 2

A clean handoff into full answer generation on Day 3

By the end of today

- HR documents are processed through the retrieval workflow
- Embeddings are generated and stored in Pinecone
- Sample user questions return relevant context
- Participants understand why retrieval is the backbone of enterprise RAG

What happens on Day 3

- Connect retrieval to the LLM
- Generate grounded responses
- Build the Gradio chatbot interface
- Demo the full DDS HR Enterprise Chatbot

Today we build the search intelligence.

Tomorrow we turn it into a real AI assistant.

DDS HR Enterprise Chatbot • AI Accelerator Bootcamp

LlamaIndex • Embeddings • Pinecone • Retrieval

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